

CHAPTER C14

MATERIAL-SPECIFIC SEISMIC DESIGN AND DETAILING REQUIREMENTS

Because seismic loading is expected to cause nonlinear behavior in structures, seismic design criteria require not only provisions to govern loading but also provisions to define the required configurations, connections, and detailing to produce material and system behavior consistent with the design assumptions. Thus, although ASCE/SEI 7-10 is primarily a loading standard, compliance with Chapter 14, which covers material-specific seismic design and detailing, is required. In general, Chapter 14 adopts material design and detailing standards developed by material standards organizations. These material standards organizations maintain complete commentaries covering their standards, and such material is not duplicated here.

C14.0 SCOPE

The scoping statement in this section clarifies that foundation elements are subject to all of the structural design requirements of the standard.

C14.1 STEEL

C14.1.1 Reference Documents. This section lists a series of structural standards published by the American Institute of Steel Construction (AISC), the American Iron and Steel Institute (AISI), the American Society of Civil Engineers (ASCE/SEI), the Steel Deck Institute (SDI), and the Steel Joist Institute (SJI), which are to be applied in the seismic design of steel members and connections in conjunction with the requirements of ASCE/SEI 7. The AISC references are available free of charge in electronic format at www.aisc.org; the AISI references are available on www.steel.org; the SDI references are available as a free download at www.sdi.org; and the SJI references are available as a free download at www.steeljoist.org.

C14.1.2 Structural Steel

C14.1.2.1 General. This section adopts AISC 360 (2016) by direct reference. The specification applies to the design of the structural steel system or systems with structural steel acting compositely with reinforced concrete. In particular, the document sets forth criteria for the design, fabrication, and erection of structural steel buildings and other structures, where other structures are defined as structures designed, fabricated, and erected in a manner similar to buildings, with buildinglike vertical and lateral load-resisting elements. The document includes extensive commentary.

C14.1.2.2 Seismic Requirements for Structural Steel Structures

C14.1.2.2.1 Seismic Design Categories B and C. For the lower Seismic Design Categories (SDCs) B and C, a range of options are available in the design of a structural steel lateral

force-resisting system. The first option is to design the structure to meet the design and detailing requirements in AISC 341 (2016) for structures assigned to higher SDCs, with the corresponding seismic design parameters (R , Ω_0 , and C_d). The second option, presented in the exception, is to use an R factor of 3 (resulting in an increased base shear), an Ω_0 of 3, and a C_d value of 3 but without the specific seismic design and detailing required in AISC 341 (2016). The basic concept underlying this option is that design for a higher base shear force results in essentially elastic response that compensates for the limited ductility of the members and connections. The resulting performance is considered comparable to that of more ductile systems.

C14.1.2.2.2 Seismic Design Categories D through F. For the higher SDCs, the engineer must follow the seismic design provisions of AISC 341 (2016) using the seismic design parameters specified for the chosen structural system, except as permitted in Table 15.4-1. For systems other than those identified in Table 15.4-1, it is not considered appropriate to design structures without specific design and detailing for seismic response in these high SDCs.

C14.1.3 Cold-Formed Steel

C14.1.3.1 General. This section adopts two standards by direct reference: ANSI/AISI S100, *North American Specification for the Design of Cold-Formed Steel Structural Members* (2016), and ASCE/SEI 8, *Specification for the Design of Cold Formed Stainless Steel Structural Members* (2002).

Both of the adopted reference documents have specific limits of applicability. ANSI/AISI S100 (2016) (Section A1.1) applies to the design of structural members that are cold-formed to shape from carbon or low-alloy steel sheet, strip, plate, or bar not more than 1 in. (25 mm) thick. ASCE/SEI 8 (2002) (Section 1.1.1) governs the design of structural members that are cold-formed to shape from annealed and cold-rolled sheet, strip, plate, or flat bar stainless steels. Both documents focus on load-carrying members in buildings; however, allowances are made for applications in nonbuilding structures, if dynamic effects are considered appropriately.

Within each document, there are requirements related to general provisions for the applicable types of steel; design of elements, members, structural assemblies, connections, and joints; and mandatory testing. In addition, ANSI/AISI S100 contains a chapter on the design of cold-formed steel structural members and connections undergoing cyclic loading. Both standards contain extensive commentaries.

C14.1.3.2 Seismic Requirements for Cold-Formed Steel Structures. This section adopts three standards by direct reference—AISI S100 (2016), ASCE/SEI 8 (2002), and AISI S400 (2015). Cold-formed steel and stainless steel members that

are part of a seismic force-resisting system listed in Table 12.2-1 must be detailed in accordance with the appropriate base standard: AISI S100 or ASCE 8.

The section also adopts a reference to AISI S400, which includes additional design provisions for a specific cold-formed steel seismic force-resisting system entitled “cold-formed steel—special bolted moment frame” or CFS-SBMF. Sato and Uang (2007) have shown that this system experiences inelastic deformation at the bolted connections because of slip and bearing during significant seismic events. To develop the designated mechanism, requirements based on capacity design principles are provided for the design of the beams, columns, and associated connections. The document has specific requirements for the application of quality assurance and quality control procedures.

C14.1.4 Cold-Formed Steel Light-Frame Construction

C14.1.4.1 General. This subsection of cold-formed steel relates to light-frame construction, which is defined as a method of construction where the structural assemblies are formed primarily by a system of repetitive wood or cold-formed steel framing members or subassemblies of these members (Section 11.2 of this standard). It adopts Section I4 of AISI S100 (2016), which directs the user to an additional suite of AISI standards, including ANSI/AISI S240 and ANSI/AISI S400.

In addition, all of these documents include commentaries to aid users in the correct application of their requirements.

C14.1.4.2 Seismic Requirements for Cold-Formed Steel Light-Frame Construction. Cold-formed steel structural members and connections in seismic force-resisting systems and diaphragms must be designed in accordance with the additional provisions of ANSI/AISI S400 in seismic design categories (SDC) D, E, or F, or wherever the seismic response modification coefficient, R , used to determine the seismic design forces is taken other than 3. In particular, this requirement includes all entries from Table 12.2-1 of this standard for “light-frame (cold-formed steel) walls sheathed with wood structural panels . . . or steel sheets,” “light-frame walls with shear panels of all other materials” (e.g., gypsum board and fiberboard panels), and “light-frame wall systems using flat strap bracing.”

C14.1.4.3 Prescriptive Cold-Formed Steel Light-Frame Construction. This section adopts ANSI/AISI S230, *Standard for Cold-Formed Steel Framing—Prescriptive Method for One and Two Family Dwellings*, which applies to the construction of detached one- and two-family dwellings, townhouses, and other attached single-family dwellings not more than two stories in height using repetitive in-line framing practices (Section A1). This document includes a commentary to aid the user in the correct application of its requirements.

C14.1.5 Cold-Formed Steel Deck Diaphragms. This section adopts the applicable standards for the general design of cold-formed steel deck diaphragms and steel roof, noncomposite floor, and composite floor deck. The SDI standards also reference ANSI/AISI S100 for materials and determination of cold-formed steel cross section strength and specify additional requirements specific to steel deck design and installation.

Additionally, design of cold-formed steel deck diaphragms is to be based on ANSI/AISI S310. All fastener design values (welds, screws, power-actuated fasteners, and button punches) for attaching deck sheet to deck sheet or for attaching the deck to the building framing members must be per ANSI/AISI S310 or specific testing prescribed in ANSI/AISI 310. All cold-formed steel deck diaphragm and fastener design properties not

specifically included in ANSI/AISI S310 must be approved for use by the authorities in whose jurisdiction the construction project occurs. Deck diaphragm in-plane design forces (seismic, wind, or gravity) must be determined per ASCE 7, Section 12.10.1. Cold-formed steel deck manufacturer test reports prepared in accordance with this provision can be used where adopted and approved by the authority having jurisdiction for the building project. The *Diaphragm Design Manual* produced by the Steel Deck Institute (2015) is also a reference for design values.

Cold-formed steel deck is assumed to have a corrugated profile consisting of alternating up and down flutes that are manufactured in various widths and heights. Use of flat sheet metal as the overall floor or roof diaphragm is permissible where designed by engineering principles, but it is beyond the scope of this section. Flat or bent sheet metal may be used as closure pieces for small gaps or penetrations or for shear transfer over short distances in the deck diaphragm where diaphragm design forces are considered.

Cold-formed steel deck diaphragm analysis must include design of chord members at the perimeter of the diaphragm and around interior openings in the diaphragm. Chord members may be steel beams attached to the underside of the steel deck designed for a combination of axial loads and bending moments caused by acting gravity and lateral loads.

Where diaphragm design loads exceed the bare steel deck diaphragm design capacity, then either horizontal steel trusses or a structurally designed concrete topping slab placed over the deck must be provided to distribute lateral forces. Where horizontal steel trusses are used, the cold-formed steel deck must be designed to transfer diaphragm forces to the steel trusses. Where a structural concrete topping over the deck is used as the diaphragm, the diaphragm chord members at the perimeter of the diaphragm and edges of interior openings must be either (a) designed flexural reinforcing steel placed in the structural concrete topping or (b) steel beams located under the deck with connectors (that provide a positive connection) as required to transfer design shear forces between the concrete topping and steel beams.

C14.1.7 Steel Cables. These provisions reference ASCE 19, *Structural Applications of Steel Cables for Buildings*, for the determination of the design strength of steel cables.

C14.1.8 Additional Detailing Requirements for Steel Piles in Seismic Design Categories D through F. Steel piles used in higher SDCs are expected to yield just under the pile cap or foundation because of combined bending and axial load. Design and detailing requirements of AISC 341 for H-piles are intended to produce stable plastic hinge formation in the piles. Because piles can be subjected to tension caused by overturning moment, mechanical means to transfer such tension must be designed for the required tension force, but not less than 10% of the pile compression capacity.

C14.2 CONCRETE

The section adopts by reference ACI 318 for structural concrete design and construction. In addition, modifications to ACI 318-14 are made that are needed to coordinate the provisions of that material design standard with the provisions of ASCE 7. Work is ongoing to better coordinate the provisions of the two documents (ACI 318 and ASCE 7) such that the provisions in Section 14.2 will be progressively reduced in future editions of ASCE 7.

C14.2.2.1 Definitions. Two definitions included here describe wall types for which definitions currently do not exist in ACI 318. These definitions are essential to the proper interpretation of the R and C_d factors for each wall type specified in Table 12.2-1.

A definition for *connector* has been added, which does not currently exist in ACI 318-14. Section 12.11 provides an alternative to the current diaphragm design procedure of Section 12.10. The alternative procedure is made mandatory for precast concrete diaphragms in structures assigned to SDC C, D, E, or F. The definition of *connector* is essential because the three design options (BDO, EDO, and RDO) are closely related to the connector classification, and the diaphragm design force reduction factor, R_s , depends on the design option.

The definition for *connection* in ACI 318-14 has also been supplemented, as it applies to this protocol.

C14.2.2.2 ACI 318, Section 10.7.6. ACI 318-14, Section 10.7.6.1.6, prescribes details of transverse reinforcement around anchor bolts in the top of a column or pedestal. This modification prescribes additional details for transverse reinforcement around such anchor bolts in structures assigned to SDCs C through F.

C14.2.2.3 Scope. This provision describes how the ACI 318-14 provisions should be interpreted for consistency with the ASCE 7 provisions.

C14.2.2.4 Intermediate Precast Structural Walls. Section 18.5 of ACI 318-14 imposes requirements on precast walls for moderate seismic risk applications. Ductile behavior is to be ensured by yielding of the steel elements or reinforcement between panels or between panels and foundations. This provision requires the designer to determine the deformation in the connection corresponding to the earthquake design displacement and then to check from experimental data that the connection type used can accommodate that deformation without significant strength degradation.

Several steel element connections have been tested under simulated seismic loading, and the adequacy of their load-deformation characteristics and strain capacity have been demonstrated (Schultz and Magana 1996). One such connection was used in the five-story building test that was part of the Precast Seismic Structural Systems (PRESSS) Phase 3 research. The connection was used to provide damping and energy dissipation, and it demonstrated a very large strain capacity (Nakaki et al. 2001). Since then, several other steel element connections have been developed that can achieve similar results (Banks and Stanton 2005 and Nakaki et al. 2005). In view of these results, it is appropriate to allow yielding in steel elements that have been shown experimentally to have adequate strain capacity to maintain at least 80% of their yield force through the full design displacement of the structure.

C14.2.2.6 Foundations. The intention is that there should be no conflicts between the provisions of ACI 318-14, Section 18.13, and ASCE 7, Sections 12.1.5, 12.13, and 14.2. However, the additional detailing requirements for concrete piles of Section 14.2.3 can result in conflicts with ACI 318-14 provisions if the pile is not fully embedded in the soil.

C14.2.2.7 Detailed Plain Concrete Shear Walls. Design requirements for plain masonry walls have existed for many years, and the corresponding type of concrete construction is the plain concrete wall. To allow the use of such walls as the lateral force-resisting system in SDCs A and B, this provision requires such walls to contain at least the minimal reinforcement specified in ACI 318-14, Section 14.6.2.2.

C14.2.3 Additional Detailing Requirements for Concrete Piles. Chapter 20 of PCI (2004) provides detailed information on the structural design of piles and on pile-to-cap connections for precast prestressed concrete piles. ACI 318-14 does not contain provisions governing the design and installation of

portions of concrete piles, drilled piers, and caissons embedded in ground except for SDC D, E, and F structures.

C14.2.3.1.2 Reinforcement for Uncased Concrete Piles (SDC C). The transverse reinforcing requirements in the potential plastic hinge zones of uncased concrete piles in SDC C are a selective composite of two ACI 318-14 requirements. In the potential plastic hinge region of an intermediate moment-resisting concrete frame column, the transverse reinforcement spacing is restricted to the least of (1) eight times the diameter of the smallest longitudinal bar, (2) 24 times the diameter of the tie bar, (3) one-half the smallest cross-sectional dimension of the column, and (4) 12 in. (304.8 mm). Outside of the potential plastic hinge region of a special moment-resisting frame column, the transverse reinforcement spacing is restricted to the smaller of six times the diameter of the longitudinal column bars and 6 in. (152.4 mm).

C14.2.3.1.5 Reinforcement for Precast Nonprestressed Piles (SDC C). Transverse reinforcement requirements inside and outside of the plastic hinge zone of precast nonprestressed piles are clarified. The transverse reinforcement requirement in the potential plastic hinge zone is a composite of two ACI 318-14 requirements (see Section C14.2.3.1.2). Outside of the potential plastic hinge region, the transverse reinforcement spacing is restricted to 16 times the longitudinal bar diameter. This restriction should permit the longitudinal bars to reach compression yield before buckling. The maximum 8-in. (203.2-mm) tie spacing comes from current building code provisions for precast concrete piles.

C14.2.3.1.6 Reinforcement for Precast Prestressed Piles (SDC C). The transverse and longitudinal reinforcing requirements given in ACI 318-14, Chapter 21, were never intended for slender precast prestressed concrete elements and result in unbuildable piles. These requirements are based on PCI Committee on Prestressed Concrete Piling (1993).

Eq. (14.2-1), originally from ACI 318-14, has always been intended to be a lower bound spiral reinforcement ratio for larger diameter columns. It is independent of the member section properties and can therefore be applied to large- or small-diameter piles. For cast-in-place concrete piles and precast prestressed concrete piles, the spiral reinforcing ratios resulting from this formula are considered to be sufficient to provide moderate ductility capacities (Fanous et al. 2007).

Full confinement per Eq. (14.2-1) is required for the upper 20 ft (6.1 m) of the pile length where curvatures are large. The amount is relaxed by 50% outside of that length in view of lower curvatures and in consideration of confinement provided by the soil.

C14.2.3.2.3 Reinforcement for Uncased Concrete Piles (SDC D through F). The reinforcement requirements for uncased concrete piles are taken from current building code requirements and are intended to provide ductility in the potential plastic hinge zones (Fanous et al. 2007).

C14.2.3.2.5 Reinforcement for Precast Nonprestressed Piles (SDC D through F). The transverse reinforcement requirements for precast nonprestressed concrete piles are taken from the IBC (ICC 2012) requirements and should be adequate to provide ductility in the potential plastic hinge zones (Fanous et al. 2007).

C14.2.3.2.6 Reinforcement for Precast Prestressed Piles (SDC D through F). The reduced amounts of transverse reinforcement specified in this provision compared with those required for special moment frame columns in ACI 318-14 are justified by the results of the study by Fanous et al. (2007). The last paragraph

provides minimum transverse reinforcement outside of the zone of prescribed ductile reinforcing.

C14.2.4 Additional Design and Detailing Requirements for Precast Concrete Diaphragms. Section 12.10.3 introduces an alternative procedure for the calculation of diaphragm design forces of Sections 12.10.1 and 12.10.2 and is made mandatory for precast concrete diaphragms in structures assigned to SDC C, D, E, or F. The diaphragm design force reduction factors, R_s , in Table 12.10-1 for precast concrete diaphragms are specifically tied to design and detailing requirements so that the ductility and overstrength necessary for expected diaphragm performance are achieved. Section 14.2.4 is based on the Diaphragm Seismic Design Methodology (DSDM), the product of a multiple-university research project termed the DSDM Project (Charles Pankow Foundation 2014), and gives detailing requirements for diaphragms constructed of precast concrete units in SDC C, D, E, or F consistent with the R_s factors. These detailing requirements are in addition to those of ACI 318, as modified by Section 14.2. The derivation of diaphragm design force reduction factors is described in Commentary Section C12.10.3.5.

Section C12.10.3.5 relates the global ductility required by the three design options defined in Section 11.2 to the local ductility of connectors measured at the maximum considered earthquake (MCE) level. The jointed nature of precast systems results in the load paths and deformations being largely determined by the connections across the joints. The connections may consist of either reinforced concrete topping slabs or discrete mechanical connectors. Since the diaphragm strains are concentrated at the joints, the connectors or the reinforcing in the topping slab must accommodate some strain demand.

C14.2.4.1 Diaphragm Seismic Demand Levels. Fig. 14.2-1 is used to determine diaphragm seismic demand level as a function of the diaphragm span and the diaphragm aspect ratio.

The diaphragm span defined in Section 14.2.4.1.1 is illustrated in Fig. C14.2-1. Most precast diaphragms contain precast units running in only one direction, and typically the maximum span is oriented perpendicular to the joints between the primary precast floor units. The connector or reinforcement deformability classifications and resulting R_s factors are calibrated relative to joint openings between the precast floor units and are thus based on the more typical orientation.

The diaphragm aspect ratio (AR) defined in Section 14.2.4.1.2 is also illustrated in Fig. C14.2-1.

The following lists provide details of seismic demand level classifications, determined in accordance with Fig. 14.2-1:

Low Seismic Demand Level

1. Diaphragms in structures assigned to SDC C.
2. Diaphragms in structures assigned to SDC D, E, or F with diaphragm span ≤ 75 ft (22.86 m), number of stories ≤ 3 , and diaphragm aspect ratio < 2.5 .

Moderate Seismic Demand Level

1. Diaphragms in structures assigned to SDC D, E, or F with diaphragm span ≤ 75 ft (22.86 m) and number of stories > 3 but ≤ 6 .
2. Diaphragms in structures assigned to SDC D, E, or F with diaphragm span > 75 ft (22.86 m) but ≤ 190 ft (57.91 m) and number of stories ≤ 2 .
3. Diaphragms in structures assigned to SDC D, E, or F with diaphragm span > 75 ft (22.86 m) but ≤ 140 ft (42.67 m) and number of stories > 2 but ≤ 4 .

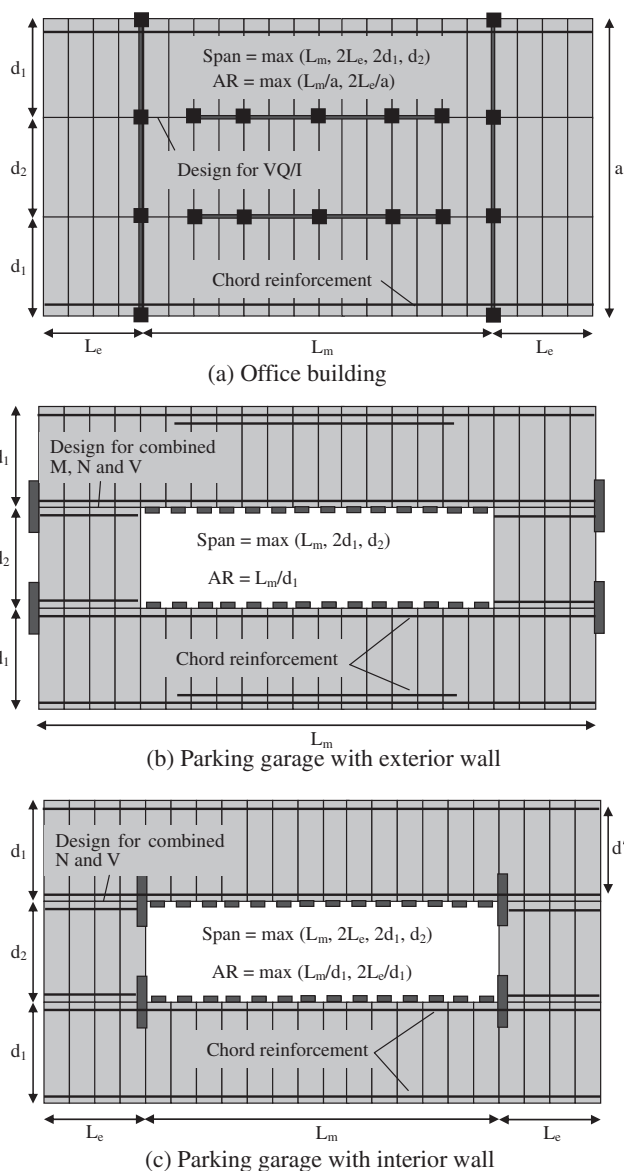


FIGURE C14.2-1 Diaphragm Dimensions

4. Diaphragms in structures assigned to SDC D, E, or F with diaphragm span ≤ 75 ft (22.86 m), number of stories ≤ 3 , and diaphragm aspect ratio ≥ 2.5 .
5. Diaphragms in structures assigned to SDC D, E, or F, categorized below as high seismic demand level, with diaphragm aspect ratio < 1.5 .

High Seismic Demand Level

1. Diaphragms in structures assigned to SDC D, E, or F with diaphragm span > 190 ft (57.91 m).
2. Diaphragms in structures assigned to SDC D, E, or F with diaphragm span > 140 ft (42.67 m) and number of stories > 2 .
3. Diaphragms in structures assigned to SDC D, E, or F with diaphragm span > 75 ft (22.86 m) and number of stories > 4 .
4. Diaphragms in structures assigned to SDC D, E, or F with number of stories > 6 .

Diaphragm Shear Overstrength Factor. The diaphragm shear overstrength factor, Ω_v , is applied to diaphragm shear reinforcement/connectors. The purpose of this factor is to keep the diaphragm shear response elastic while the diaphragm develops inelastic flexural action, as is anticipated for the basic design objective (BDO) in the MCE, and for the reduced design objective (RDO) for both the design earthquake and the MCE. No inelastic diaphragm response is anticipated for the elastic design objective (EDO).

The value of diaphragm shear overstrength factor is $\Omega_v = 1.4R_s$. The values of the diaphragm design force reduction factor, R_s , are 0.7, 1.0, and 1.4 for the EDO, BDO, and RDO, respectively. This value translates into diaphragm shear overstrength factors Ω_v of 1.0, 1.4, and 2.0 (rounded to one decimal place) for the EDO, BDO, and RDO, respectively.

The diaphragm shear overstrength factor, Ω_v , is applied to the diaphragm design forces and thus is a measure relative to the flexural strength of the diaphragm. As implied by the above-listed Ω_v values, the level of overstrength required relative to the diaphragm flexural strength varies with the design option. The RDO requires a higher overstrength than the BDO because of the larger anticipated inelastic action. For the EDO, no overstrength is required since the diaphragm design force itself targets elastic behavior in the MCE. It is noted that the absolute shear strength required in the design procedure is constant, regardless of design option, since the parameter R_s in the overstrength factor is canceled out by the R_s in the denominator of the diaphragm design force expression.

The Ω_v values represent upper bound constant values (for each diaphragm design objective) of parametric expressions developed for the required shear overstrength on the basis of detailed parametric studies performed using nonlinear dynamic time history analysis (NTHA) of analytical models of precast structures developed and calibrated on the basis of extensive large-scale physical testing. These precast structures were subjected to spectrum compatible ground motions scaled to the MCE in order to determine the required shear overstrength factors.

Precast diaphragms can be designed and detailed for ductile flexural response. However, to achieve the desired mechanism, potentially nonductile shear limit states have to be precluded. In order to prevent these shear failures, elastic shear response is targeted in the design procedure for both flexure-controlled and shear-controlled systems. Thus, the shear overstrength factor, Ω_v , is applied in diaphragm shear design.

The shear amplification factor values were obtained by bounding the maximum shear force $V_{\max}$ occurring in NTHA of the diaphragm at the critical shear joint as the diaphragm developed a flexural mechanism (in other regions of the floor) at MCE-level hazard and scaling it by the design shear, V_u . Accordingly:

- Ω_E , the diaphragm shear amplification factor for the EDO, is taken as unity ($\Omega_E = 1.0 \approx 1.4R_s$, where $R_s = 0.7$ for

EDO) since elastic diaphragm response is expected in the MCE for EDO.

- Ω_B , the diaphragm shear amplification factor for the BDO, is taken as an upper bound on the $V_{\max}/V_u$ ratio for the BDO design under MCE-level hazard.
- Ω_R , the diaphragm shear amplification factor for the RDO, is taken as an upper bound on the $V_{\max}/V_u$ ratio for the RDO design under MCE-level hazard.

Fig. C14.2-2 shows a scatter plot of the $V_{\max}/V_u$ ratios from NTHA for different numbers of stories (n) and diaphragm aspect ratios (ARs) at the maximum considered earthquake. The data represent the mean of the maximum responses from five ground motions. The expression provided for Ω_v , $\Omega_v = 1.4R_s$, is plotted as a horizontal dashed line on each plot, indicating that the expression provides a constant upper bound for the anticipated required elastic shear forces for all design cases.

C14.2.4.2 Diaphragm Design Options. The intent of the design procedure is to provide the diaphragm with the proper combination of strength and deformation capacity in order to survive anticipated seismic events. Three different design options are provided to the designer to accomplish this objective, ranging from a fully elastic diaphragm design under the MCE to designs that permit significant inelastic deformations in the diaphragm under the design earthquake. The motivation for this approach is the recognition that, under certain conditions, a precast diaphragm designed to remain fully elastic up to the MCE may not be economical or reliable. Under other conditions, however, a diaphragm designed to remain elastic up to the MCE will be satisfactory and may be most desirable.

The methodology allows the designer three options related to deformation capacity:

1. An elastic design option (EDO), where the diaphragm is designed to the highest force levels, is calibrated to keep the diaphragm elastic not only for the design earthquake but also in an MCE. In exchange for the higher design force, this option permits the designer to detail the diaphragm with the low deformability element (LDE) connector or reinforcement that need not meet any specific deformation capacity requirements (tension deformation capacity less than 0.3 in. (7.6 mm). This option is limited in its use through the introduction of diaphragm seismic demand levels, which are based on building height, diaphragm geometry, and seismic hazard level. The use of the EDO is not permitted if the diaphragm seismic demand level is high.
2. A basic design objective (BDO) is one in which the diaphragm is designed to a force level calibrated to keep the diaphragm elastic in the design earthquake but not necessarily in the MCE. The design force level is lower than that required for the EDO, but this option requires moderate deformability element (MDE) connectors or

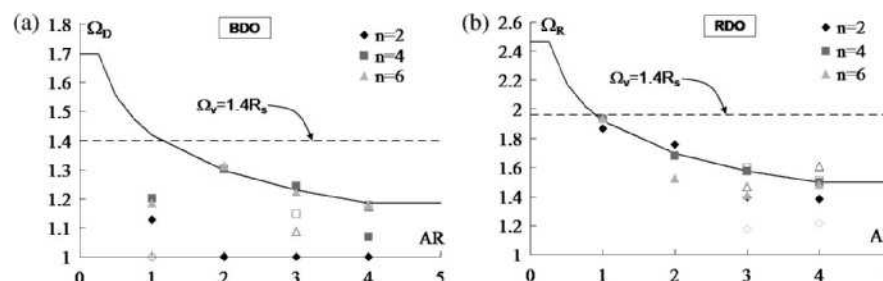


FIGURE C14.2-2 Diaphragm Shear Amplification Factor Results from NTHA at MCE: (a) BDO; (b) RDO

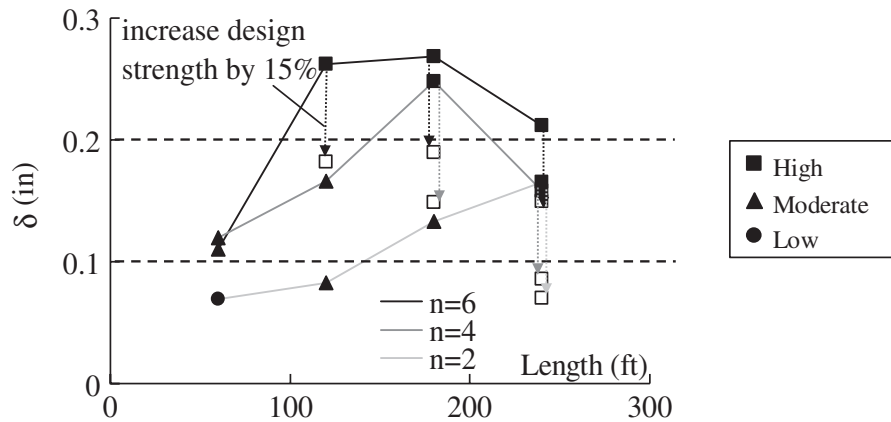


FIGURE C14.2-3 Diaphragm Maximum Joint Opening in NTHA for Basic Design Objective Designs under the MCE

reinforcement or better to provide an inelastic deformation capacity sufficient to survive the anticipated deformation demands in an MCE. This option and the RDO require the use of a diaphragm shear overstrength factor, Ω_v , to ensure that a nonductile shear failure does not occur before the connectors or reinforcement reaches its intended inelastic deformation. Note that *inelastic deformation* is associated with joint *opening* caused by diaphragm flexure, not joint sliding deformation caused by shear.

3. A reduced design option (RDO) is one in which the diaphragm is designed for the lowest design force level.

Because the design force level is lower than in the BDO, some yielding in the diaphragm is anticipated in the design earthquake. The force levels have been calibrated to keep diaphragm inelastic deformation demands in an MCE within the allowable deformation capacity for the high deformability element (HDE), the highest classification of precast diaphragm connector or reinforcement (see Section 14.2.4.3).

Each design option can be used with its associated seismic demand level or a lower seismic demand level. A 15% diaphragm force increase penalty is applied when a diaphragm design option is used for a seismic demand level that is one higher than its associated seismic demand level. A design option cannot be used for a seismic demand level two higher than the associated seismic demand level, i.e., the elastic design option cannot be used for the high seismic demand level.

The BDO has two performance targets: (1) elastic diaphragm response in the design earthquake, and (2) diaphragm connector/reinforcement deformation demands (i.e., joint opening) in the MCE within the allowable deformation capacity of connector/reinforcement in the moderate deformability element (MDE) category, δ_a^{MD} . The diaphragm design force levels for the BDO are aligned to the former requirement. Thus, the attainment of the second performance target hinges on the selection of the value for δ_a^{MD} relative to the diaphragm inelastic deformation demands anticipated for the maximum considered earthquake. These anticipated deformation demands were established through nonlinear dynamic time history analysis (NTHA) of precast structures with diaphragms designed to the BDO force levels and subjected to spectrum compatible ground motions scaled to the MCE.

It should be recognized that practical considerations also exist in the selection of δ_a^{MD} . The allowable deformation of high deformability elements (HDEs), δ_a^{HD} , (as required for the RDO) was established based on the best performing existing precast diaphragm connectors. This performance resulted in an HDE allowable deformation capacity $\delta_a^{HD} = 0.4$ in. ($\delta_a^{HD} = 10.2$ mm).

(Note that the allowable value is 2/3 of the qualification value, thus HDEs are required to have a demonstrated deformation capacity of 0.6 in. (15.2 mm) in qualification testing, as was achieved by the best performing existing connectors). Given that low deformability elements (LDEs) do not have a deformation requirement, the MDE allowable deformation value should reside somewhere near half the HDE value, or $\delta_a^{MD} = 0.2$ in. ($\delta_a^{MD} = 5.1$ mm).

The NTHA results for the MCE are shown in Fig. C14.2-3. These results show that $\delta_a^{MD} = 0.2$ in. ($\delta_a^{MD} = 5.1$ mm) was an appropriate and viable choice for the MDEs used in the BDO, provided that the diaphragms were in the moderate seismic demand level (solid triangles in Fig. C14.2-3) or in the low seismic demand level (solid circles in Fig. C14.2-3). However, this value did not produce satisfactory designs for diaphragms in the high seismic demand level (solid squares in Fig. C14.2-3), and thus some measure is required to bring the design procedure in conformance.

A choice exists in how to modify the design procedure to resolve this nonconformance to the design target: (a) The allowable deformation ranges for the diaphragm connectors/reinforcement could be modified (i.e., a more stringent qualification deformation requirement for MDE, leading to an increase in δ_a^{MD}); (b) the diaphragm force levels could be increased across the board (i.e., change the design earthquake performance target for elastic diaphragm response from the diaphragm yield point itself to a lower value within the diaphragm elastic range); or (c) create a special requirement for the nonconforming diaphragm case (i.e., increase the diaphragm forces only for nonconforming cases). The first choice did not align well with the typical deformation capacities of existing connectors and would not produce evenly sized deformation ranges for the LDE, MDE, and HDE classifications. The second choice not only produces overly conservative designs for many cases, but it also blurs the clean BDO performance target of elastic diaphragm response in the design earthquake. For these reasons, the third choice was considered the most desirable.

Thus, rather than increase the value of δ_a^{MD} to accommodate the diaphragms in the high seismic demand level, it was decided to keep $\delta_a^{MD} = 0.2$ in. ($\delta_a^{MD} = 5.1$ mm) and create a special requirement for conformance in the case of diaphragms in the high seismic demand level. As each design option was developed with an associated diaphragm seismic demand level in mind, and the nonconformance did not occur at the associated level, i.e., the moderate seismic demand level, but instead at the high seismic demand level, the special requirement can be considered a measure for using a diaphragm design option with a more demanding seismic demand level.

The special requirement is an increase in the design force for the nonconforming case. The magnitude of the design force increase is 15%. The manner in which this value was established is also shown in Fig. C14.2-3. As mentioned previously, the solid squares indicate the maximum diaphragm connector/reinforcement deformation (joint opening demand) for the BDO for high diaphragm seismic demand levels and indicate demands greater than $\delta_a^{MD} = 0.2$ in. (5.1 mm). The open squares indicate the maximum diaphragm connector/reinforcement deformation for these same cases with the 15% increase in diaphragm force. This design force increase is seen to bring the deformation demand within the allowable limit. The same design force increase is enforced in Section 14.2.4.2.1 for use of the EDO with the moderate seismic demand level, though this provision was not based on any quantitative analytical results.

C14.2.4.3 Diaphragm Connector or Joint Reinforcement Deformability. The precast diaphragm seismic design methodology (DSDM) uses an approach that requires knowledge of the diaphragm connector or reinforcement stiffness, deformation capacity, and strength to effectively and efficiently design the diaphragm system for seismic forces. To meet this need, it is critical that the connector or reinforcement properties be determined in a repeatable, reproducible, and consistent manner so that existing and new connections can be used effectively in the diaphragm system. The qualification procedure provides an experimental approach for the determination of connector or reinforcement properties.

Precast concrete diaphragms deform mostly by the strains that occur at the joints between the precast concrete units. The requirements for reinforcement or connector deformability come from the need for the connections to accommodate these strains at the joints. A connection is an assembly of connectors, including the linking parts, welds, and anchorage to concrete. Mechanical connectors are identified as the primary parts that make the connection, but the deformation capacity identified with the connector represents the performance of the entire link across the joint. Qualification of the deformation capacity of the connector, then, is dependent on the details of the entire load path across the joint. The use in design of a connector qualified by testing is only valid when the design incorporates the complete connector detailing, as tested.

The diaphragm reinforcement classifications are high deformability elements (HDEs), moderate deformability elements (MDEs), and low deformability elements (LDEs). The threshold values of tension deformation capacity for each connector or reinforcement class were selected by considering the range of the ultimate (cyclic tension opening) deformations exhibited by the various precast diaphragm connectors examined in the DSDM experimental program (Naito et al. 2006, 2007). Based on these results, a threshold deformation of 0.6 in. (15.2 mm) was selected for HDE connector or reinforcement and 0.3 in. (7.6 mm) for MDE connector or reinforcement. There is no deformation requirement for LDE reinforcement.

A factor of safety of 1.5 was introduced into the design procedure by establishing the allowable maximum joint opening value at 2/3 of the connector's reliable and maximum joint opening deformation capacity. The 2/3 factor leads to maximum allowable deformations of 0.4 in. (10.2 mm) and 0.2 in. (5.1 mm) for the high deformability element (HDE) and the moderate deformability element (MDE), respectively. No deformation capacity requirement is needed for the low deformability element (LDE), since this classification of connector or reinforcement is used with designs that result in fully elastic diaphragm response up to the MCE. The allowable

maximum joint openings were used as targets in the analytical parametric studies to calibrate the design factors.

A few further comments are given about the connector or reinforcement classification:

1. The diaphragm connector or reinforcement classification is based on inelastic deformation associated with joint opening caused by diaphragm flexure, not joint sliding deformation caused by shear.
2. The diaphragm connector or reinforcement classification applies to the chord reinforcement and shear reinforcement. Other reinforcement (collector/anchorage, secondary connections to spandrels, and similar items) may have different requirements or characteristics.
3. In meeting the required maximum deformation capacity using the testing protocols in the qualification procedure, the required cumulative inelastic deformation capacity is also met.

C14.2.4.3.5 Deformed Bar Reinforcement. Deformed bar reinforcement can be considered to be high deformability elements (HDEs), provided that certain conditions are met.

C14.2.4.3.6 Special Inspection. The purpose of this requirement is to verify that the detailing required in HDEs is properly executed through inspection personnel who are qualified to inspect these elements. Qualifications of inspectors should be acceptable to the jurisdiction enforcing the general building code.

C14.2.4.4 Precast Concrete Diaphragm Connector and Joint Reinforcement Qualification Procedure. This section provides a qualification procedure using experimental methods to assess the in-plane strength, stiffness, and deformation capacity of precast concrete diaphragm connectors and reinforcement. The methodology was developed as part of the DSDM research program specifically for diaphragm flange-to-flange connections and is intended to provide the required connector or reinforcement properties and classification for use in the seismic design procedure.

C14.2.4.4.1 Test Modules. Test modules are fabricated and tested to evaluate the performance of a precast concrete connection. Fig. C14.2-4 illustrates an example test module. It is required that multiple tests be conducted to assess repeatability and consistency. The test module should represent the geometry and thickness of the precast concrete components that will be connected. All connectors and reinforcement should be installed and welded in accordance with the manufacturer's published installation instructions. The results or the data generated are limited to connections built to the specified requirements.

Reduced scale connectors with appropriate reductions in maximum aggregate size following laws of similitude can be

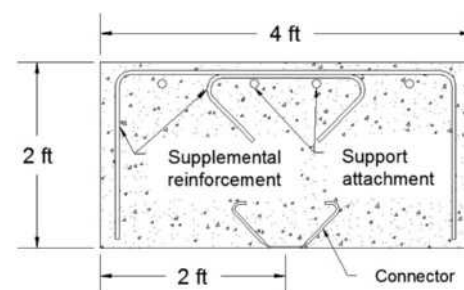


FIGURE C14.2-4 Test Module

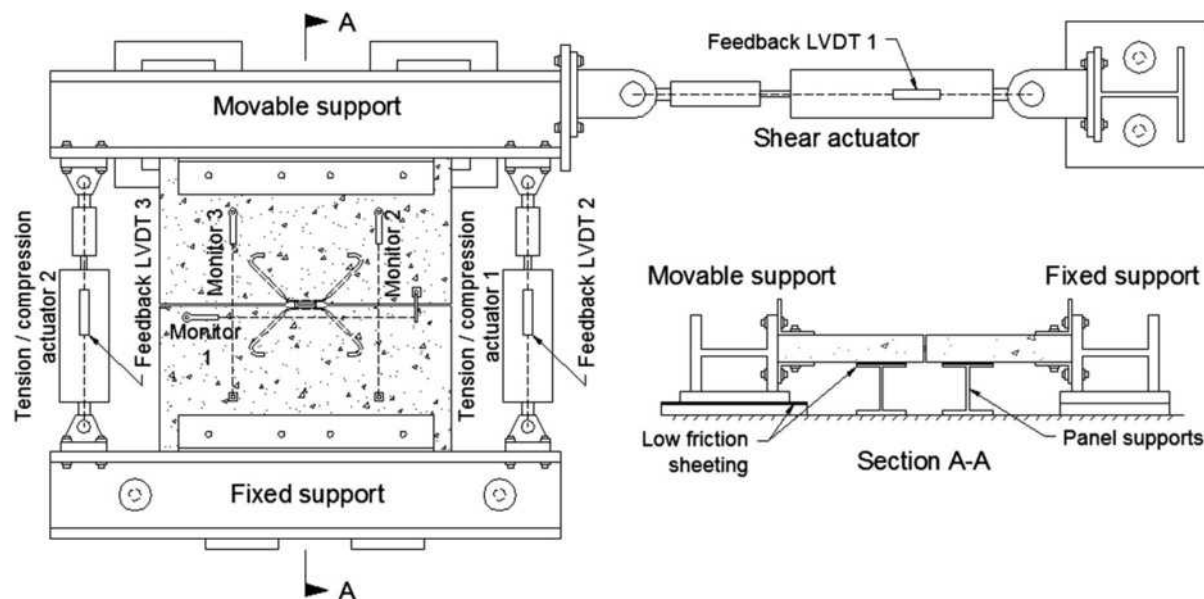


FIGURE C14.2-5 Possible Test Setup

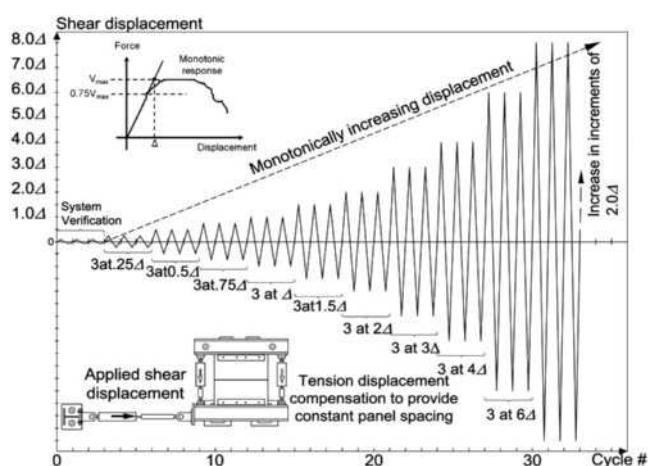


FIGURE C14.2-6 Shear Loading Protocol

used as research tools to gain knowledge but are not to be used for connector qualification.

C14.2.4.4.3 Test Configuration. A possible setup is illustrated in Fig. C14.2-5. Three independently controlled actuators are used, two providing axial displacement and one providing shear displacement to the connection.

C14.2.4.4.4 Instrumentation. Use of actuator transducers is not recommended because of potential slip in the test fixture.

C14.2.4.4.5 Loading Protocols. Figs. C14.2-6 and C14.2-7 illustrate the shear and tension/compression loading protocols for use in testing.

C14.2.4.4.6 Measurement Indices, Test Observations, and Acquisition of Data. Quantitative data should be recorded from each test, such that interpretation can be made of the performance of the test module. For in-plane tests, the axial and shear force and deformations should be recorded. Photographs should be taken to illustrate the condition of the test module at the initiation

and completion of testing as well as at points through the testing history. Ideally, photos should be taken at the end of each group of cycles. Test history photos taken at points of interest, such as cracking, yielding, and peak load, and post test photos are adequate for most evaluations.

The backbone curve is adopted to represent a simple approximation of the load-deformation response of the connection. The points are defined in terms of the resistances P_a , P_1 , P_b , P_2 , P_{2a} , and P_3 , and the displacements Δ_a , Δ_1 , Δ_b , Δ_2 , Δ_{2a} , and Δ_3 , respectively.

As depicted in Fig. 14.2-3, the Type 1 curve is representative of ductile behavior where there is an elastic range (Point 0 to Point 1 on the curve) and an inelastic range (Point 1 to Point 3 on the curve), followed by loss of force-resisting capacity. The Type 2 curve is representative of ductile behavior where there is an elastic range (Point 0 to Point 1) and an inelastic range (Point 1 to Point 2 on the curve), followed by substantial loss of force-resisting capacity. Some connections may exhibit a small peak strength with limited ductility. For these cases, the Alternate Type 2 curve is recommended. The Type 3 curve is representative of a brittle or nonductile behavior where there is an elastic range (Point 0 to Point 1) followed by loss of strength. Deformation-controlled elements conform to Type 1 or Type 2, but not Type 2 Alternate, response with $\Delta_2 \geq 2\Delta_1$. All other responses are classified as force-controlled. An example of test data is included in Ren and Naito (2013).

C14.2.4.4.7 Response Properties. The reliable and stable maximum deformation capacity is defined for design code purposes as the connector deformation at peak load, Point 2 on the backbone curve, obtained in testing following the loading protocols defined here. All analytical calibrations were performed for a reliable and stable maximum deformation capacity corresponding to a deformation where the strength reduces to 80% of P_2 , which is similar to the beam-column connection deformation capacity definition for steel structures in AISC 341. Thus, an added degree of conservatism is provided in the definition proposed for the design code.

Deformation Category. The category ranges were determined from finite element analysis of a database of diaphragm systems under a range of seismic demands. Alternate deformation limits

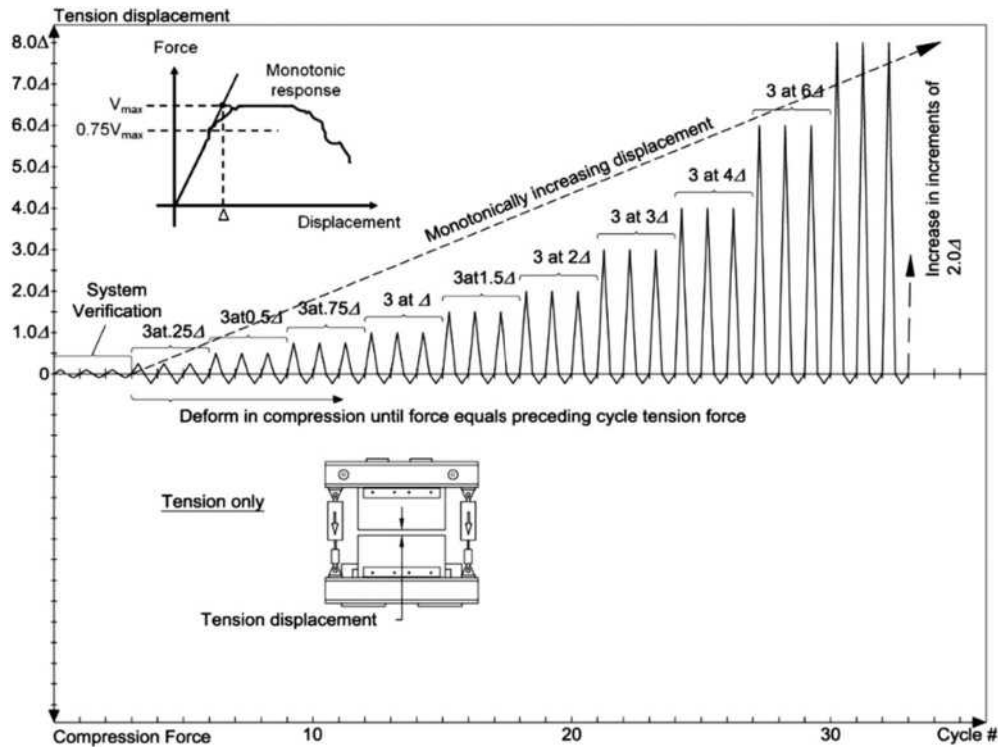


FIGURE C14.2-7 Tension/Compression Loading Protocol

can be used if supporting data are provided. It should be noted that the connector or joint reinforcement classification is based solely on *tension* deformation capacity (as stated in Section 14.2.4.3), whereas the qualification procedure applies equally to, and requires, both tension and shear tests. In other words, while both tension and shear characterization are required to determine the needed strengths, the connector classification is based solely on the tension testing.

Tensile Strength. The design factors for flexural strength are calibrated to the yield point of the chord connectors, not to their peak strength. For instance, for the EDO, elastic response of the diaphragm under the MCE is being targeted, so this response is aligned to the yield strength, not the peak strength. For consistency, the BDO and RDO factors are also calibrated to this same level, i.e., yield. So the nominal strength of the connectors is based on P_1 , not P_2 . Using P_2 creates a situation where yield should be anticipated in the diaphragm for the EDO, and larger inelastic deformations for the BDO and RDO.

Shear Strength. The intention is for the diaphragm system to remain elastic under shear demands. Consequently, the inelastic shear force capacity of connections is not considered. Because of the existence of low stiffness connections, limits are placed on the allowable deformation at which the force P_1 can be determined.

C14.2.4.4.8 Test Report. The minimum information that must be included in a test report is spelled out.

C14.3 COMPOSITE STEEL AND CONCRETE STRUCTURES

This section provides guidance on the design of composite and hybrid steel–concrete structures. Composite structures are defined as those incorporating structural elements made of steel and concrete portions connected integrally throughout the structural element by mechanical connectors, bonds, or both. Hybrid

structures are defined as consisting of steel and concrete structural elements connected together at discrete points. Composite and hybrid structural systems mimic many of the existing steel (moment and braced frame) and reinforced concrete (moment frame and wall) configurations but are given their own design coefficients and factors in Table 12.2-1. Their design is based on ductility and energy dissipation concepts comparable to those used in conventional steel and reinforced concrete structures, but it requires special attention to the interaction of the two materials because it affects the stiffness, strength, and inelastic behavior of the members, connections, and systems.

C14.3.1 Reference Documents. Seismic design for composite structures assigned to SDCs D, E, or F is governed primarily by AISC 341. Composite design provisions in ANSI/AISC 341 are less prescriptive than those for structural steel and provide flexibility for designers to use analytical tools and results of research in their practice. Composite structures assigned to SDC A, B, or C may be designed according to principles outlined in AISC 360 and ACI 318. ANSI/AISC 360 and ACI 318 provide little guidance on connection design; therefore, designers are encouraged to review ANSI/AISC 341 for guidance on the design of joint areas. Differences between older AISC and ACI provisions for cross-sectional strength for composite beam–columns have been minimized by changes in the latest AISC 360, and AISC 360 refers to ACI 318 for much of the design of reinforced concrete components of composite structures. However, there is not uniform agreement between the provisions in ACI 318 and AISC 360 regarding detailing, limits on material strengths, stability, and strength for composite beam–columns. The composite design provisions in ANSI/AISC 360 are considered to be current.

C14.3.4 Metal-Cased Concrete Piles. Design of metal-cased concrete piles, which are analogous to circular concrete filled

tubes, is governed by Sections 14.2.3.1.3 and 14.2.3.2.4 of this standard. The intent of these provisions is to require metal-cased concrete piles to have confinement and protection against long-term deterioration comparable to that for uncased concrete piles.

C14.4 MASONRY

This section adopts by reference and then makes modifications to TMS 402 and TMS 602. In past editions of this standard, modifications to the TMS referenced standards were also made. During the development of the 2016 edition of TMS standards, each of these modifications was considered by the TMS 402/602 committee. Some were incorporated directly into the TMS standards. These modifications have accordingly been removed from the modifications in this standard. Work is ongoing to better coordinate the provisions of the two documents so that the provisions in Section 14.4 are significantly reduced or eliminated in future editions.

C14.5 WOOD

C14.5.1 Reference Documents. Two national consensus standards are adopted for seismic design of engineered wood structures: the *National Design Specification* (AWC NDS-15), and the *Special Design Provisions for Wind and Seismic* (AWC SDPWS-15). Both of these standards are presented in dual allowable stress design (ASD) and load and resistance factor design (LRFD) formats. Both standards reference a number of secondary standards for related items such as wood materials and fasteners. AWC SDPWS addresses general principles and specific detailing requirements for shear wall and diaphragm design and provides tabulated nominal unit shear capacities for shear wall and diaphragm sheathing and fastening. The balance of member and connection design is to be in accordance with the AWC NDS.

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